DPCN Research Mapping · Group 29

Research output, citation and collaboration networks of four CSIR biology labs, 2016–2026

CSIR-CCMB Hyderabad · CSIR-IGIB New Delhi · CSIR-IMTECH Chandigarh · CSIR-CDRI Lucknow

6,063	144,139	274	66%	123
unique papers	citations	faculty with papers	Q1 share of ranked journal papers	institutional h-index

Source: OpenAlex snapshot 2026-10-13; Scimago SJR; CORE2023. Instructor: Chittaranjan Hens. 2026 is a partial year.

Data and method

A reproducible OpenAlex pipeline; every number here is regenerated from the submitted CSVs

1 Roster

294 scientists: lab websites plus PIs found from lab-affiliated authorship

2 Match

Each scientist matched to one or more OpenAlex author IDs (ORCID, own publication list, name)

3 Filter

A paper counts only if the scientist's own authorship carries the lab; articles, reviews, letters, conference papers

4 Enrich

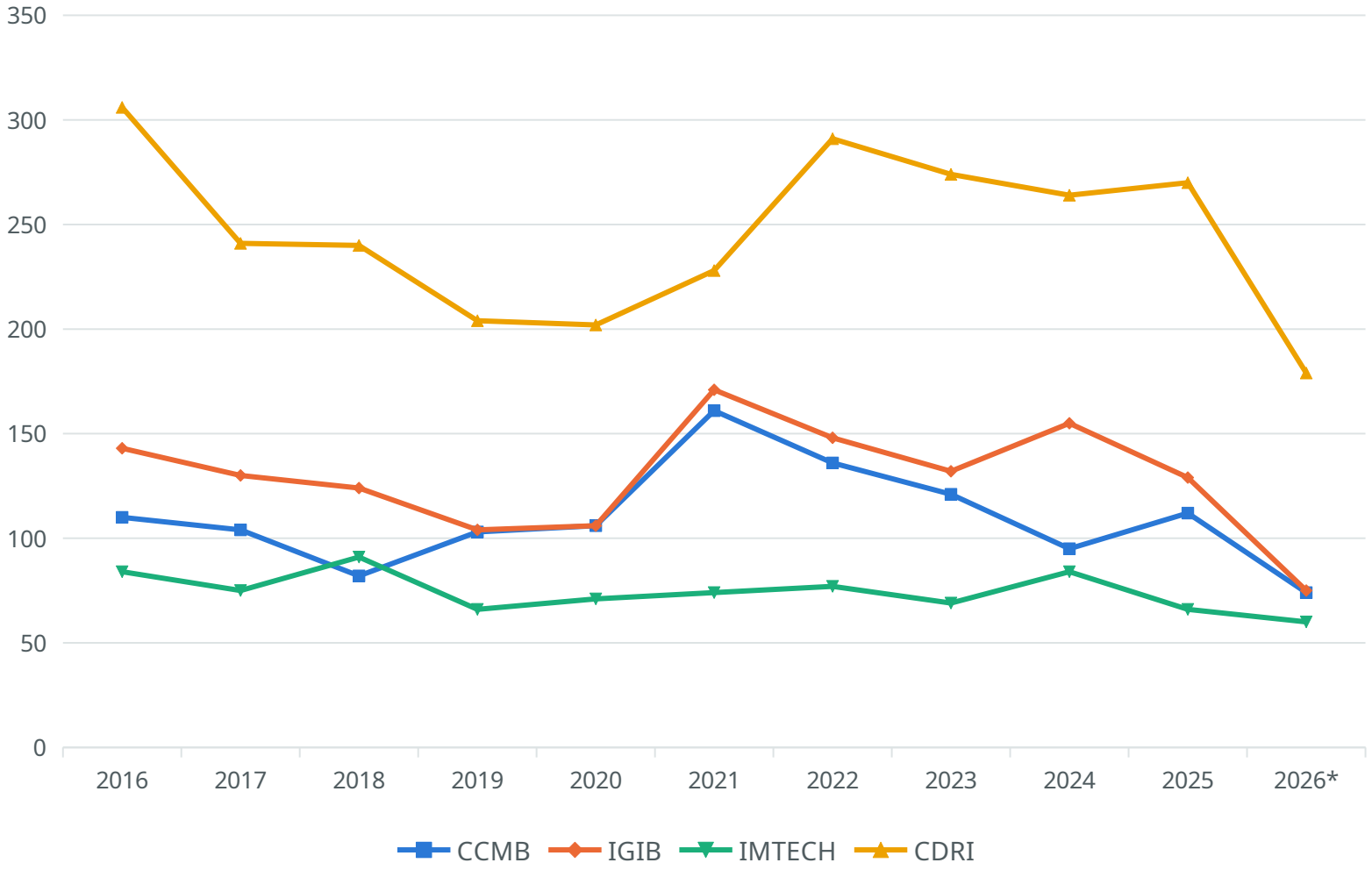
SJR quartile of the publication year (coverage 93%–98%); OpenAlex topics; Indian citation flows

Lab	Faculty	Papers	Coverage of lab output
CCMB	54	1,204	71%
IGIB	55	1,417	74%
IMTECH	49	817	72%
CDRI	116	2,699	80%

- Faculty = scientists/PIs; students and postdocs appear only as co-authors
- h-index is computed within 2016–2026 from this dataset (lifetime h shown for context)
- Indian partner = an Indian institution other than the paper's own lab (CSIR HQ, ministries, AcSIR excluded)
- Limitations: OpenAlex author disambiguation; uncovered papers are mostly by students or unlisted staff

Publication trend 2016–2026

CDRI publishes most · peak years: CCMB 2021, IGIB 2021, IMTECH 2018, CDRI 2016 · 2026 is a partial year



CCMB

1,204 papers · 36,616 citations

CAGR 2016–2025: 0.2%

IGIB

1,417 papers · 36,787 citations

CAGR 2016–2025: -1.1%

IMTECH

817 papers · 22,901 citations

CAGR 2016–2025: -2.6%

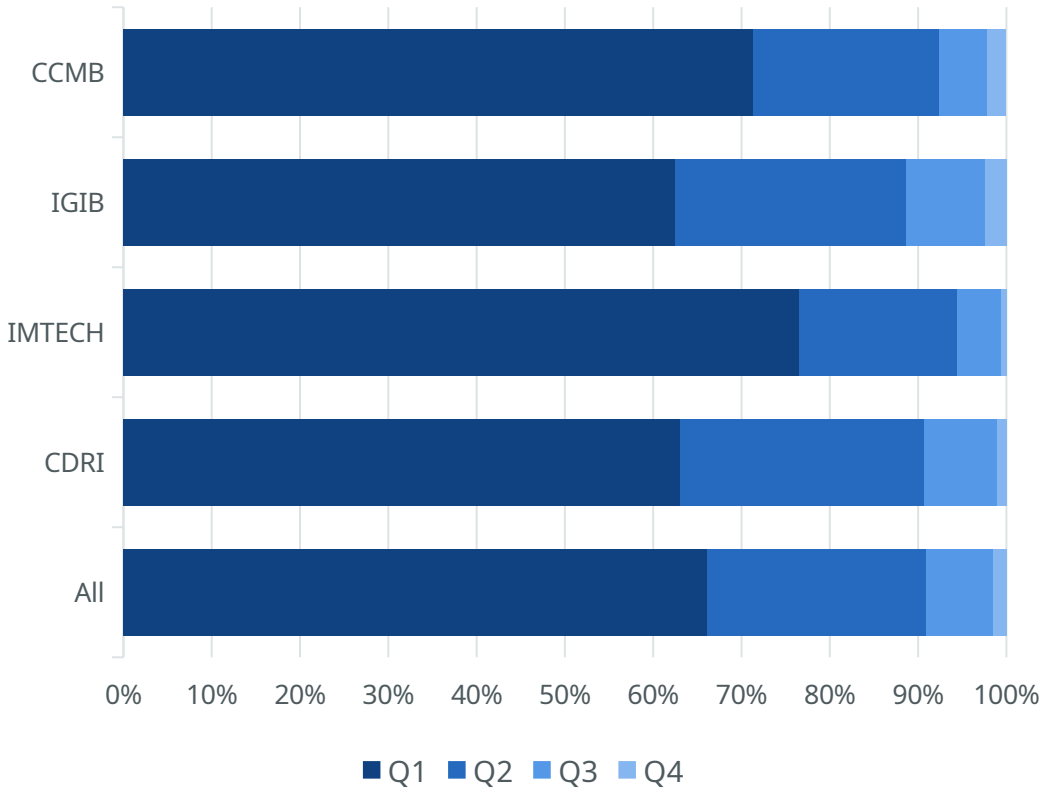
CDRI

2,699 papers · 55,256 citations

CAGR 2016–2025: -1.4%

Journal quality: Q1 fraction and top journals

Q1 fraction = Q1 papers ÷ journal papers with a known SJR quartile



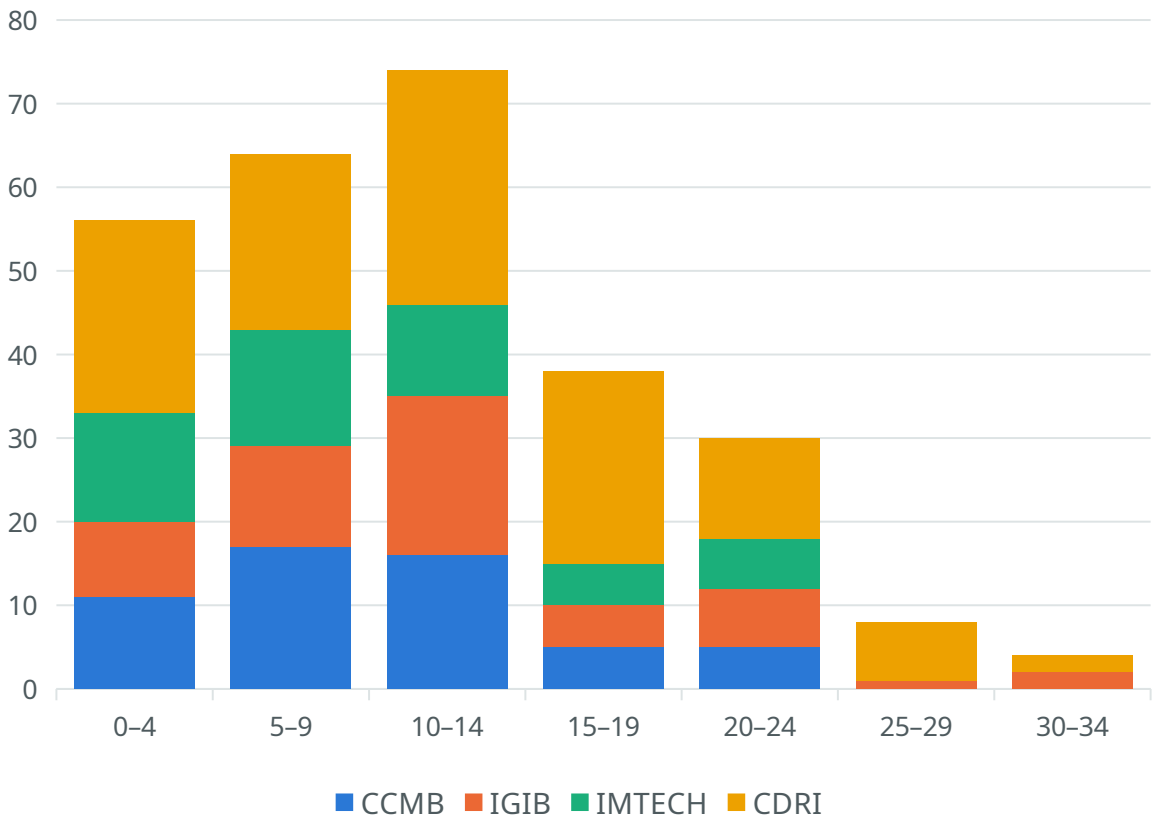
CCMB Q1 71% (coverage 93%) IGIB Q1 62% (coverage 95%) IMTECH Q1 76% (coverage 98%) CDRI Q1 63% (coverage 97%)

Top 3 journals per lab (top 5 in the dashboard)

Lab	Journal	Papers	Q
CCMB	Scientific Reports	54	Q1
	eLife	20	Q1
	Journal of Biosciences	19	Q1
IGIB	Scientific Reports	54	Q1
	Indian Journal of Microbiology	21	Q3
	Frontiers in Microbiology	21	Q1
IMTECH	Scientific Reports	47	Q1
	International Journal Of Systematic And…	38	Q1
	Frontiers in Microbiology	33	Q1
CDRI	European Journal of Medicinal Chemistry	59	Q1
	Organic Letters	55	Q1
	The Journal of Organic Chemistry	46	Q1

Faculty: h-index distribution and leaders

Window h-index from 2016–2026 dataset papers; department-level summary from the roster divisions



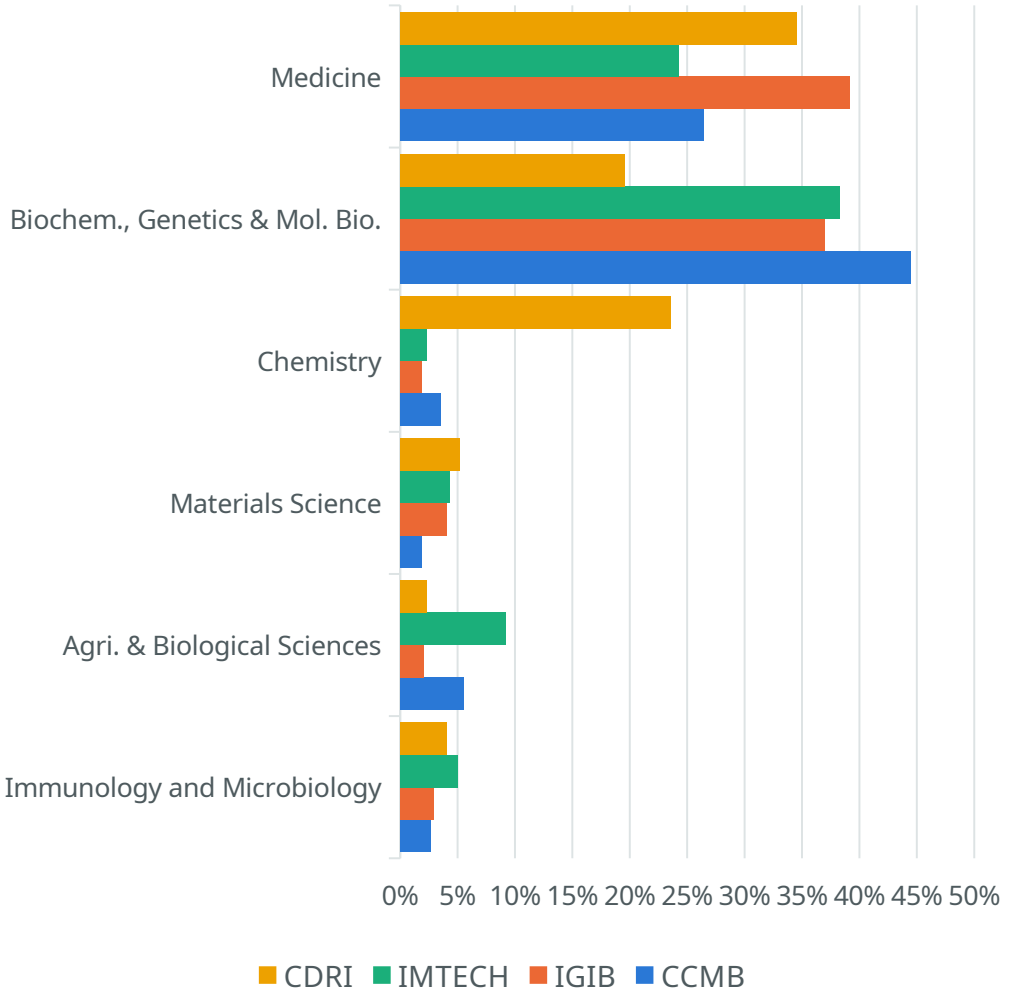
Median h: CCMB 9 · IGIB 11 · IMTECH 8 · CDRI 12 · all 10

Faculty	Lab	Papers	Citations	h
Rajesh Pandey	IGIB	137	4,853	32
Vinod Scaria	IGIB	160	4,262	31
Naibedya Chattopadhyay	CDRI	125	3,042	31
Jiaur R Gayen	CDRI	132	2,950	31
Kalyan Mitra	CDRI	85	1,832	29
Rajender Singh	CDRI	97	2,711	28
Prabhat R Mishra	CDRI	59	1,664	28
Sarika Singh	CDRI	66	2,248	27
Sidharth Chopra	CDRI	185	2,516	26
Muhammad Wahajuddin	CDRI	66	2,088	26

- 108 departments/divisions; largest by output: Medicinal & Process Chemistry (CDRI, 487 papers)
- IMTECH publishes no divisions: grouped by each scientist's main OpenAlex subfield

Research areas and most-cited papers

OpenAlex primary topic → field; citation counts at collection date



#	Paper	Journal	Year	Cites	Lab
1	Guidelines for the use and interpretation of assays for monitoring…	Autophagy	2016	5,860	CCMB
2	Guidelines for the use and interpretation of assays for monitoring…	Autophagy	2021	2,611	IGIB; IMTECH
3	Oxidative Stress: A Key Modulator in Neurodegenerative Diseases	Molecules	2019	2,251	IGIB
4	The Simons Genome Diversity Project: 300 genomes from 142 diverse…	Nature	2016	1,595	CCMB
5	Nanoemulsion: Concepts, development and applications in drug delivery	Journal of Controlled…	2017	1,354	CDRI
6	SARS-CoV-2 B.1.617.2 Delta variant replication and immune evasion	Nature	2021	1,337	CCMB; IGIB
7	Altered TMPRSS2 usage by SARS-CoV-2 Omicron impacts infectivity and…	Nature	2022	1,148	IGIB
8	The genetic architecture of type 2 diabetes	Nature	2016	1,049	CCMB; IGIB
9	The power of genetic diversity in genome-wide association studies of…	Nature	2021	968	CCMB; IGIB
10	A saturated map of common genetic variants associated with human height	Nature	2022	815	CCMB

Citation network within the four labs

Directed graph: nodes = dataset papers, edges = paper A cites paper B (internal corpus)

6,063
papers (nodes)

9,634
internal citations

56%
in largest component

49%
never cited internally

cites →	CCMB	IGIB	IMTECH	CDRI
CCMB	1659	69	35	28
IGIB	85	1542	27	31
IMTECH	30	30	1192	30
CDRI	47	46	50	5035

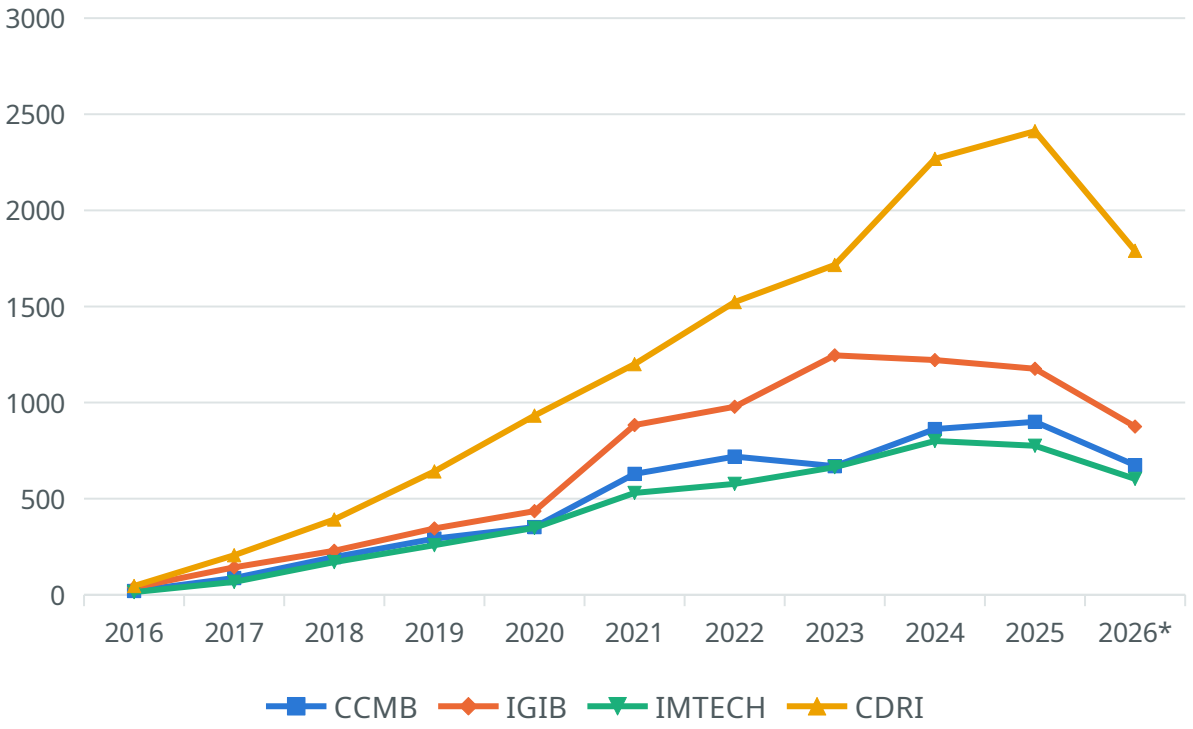
- Mean in-degree 1.59; maximum 31; reciprocity 0.006
- Heavy-tailed: the most-cited 10% of papers receive 51% of internal citations
- 95% of lab-to-lab citations stay inside the citing lab (matrix diagonal)

PageRank top 10

Paper	Year	Lab	In-deg
Nitric oxide-mediated apoptosis of neutrophils through…	2016	CDRI	12
Theophylline, a methylxanthine drug induces osteopenia and…	2016	CDRI	23
Altered glucose and lipid homeostasis in liver and adipose…	2017	CDRI	13
Computational Analysis and Phylogenetic Clustering of…	2021	IGIB	6
The Simons Genome Diversity Project: 300 genomes from 142…	2016	CCMB	12
High throughput detection and genetic epidemiology of…	2021	IGIB	15
An F-type lectin domain directs the activity of…	2018	IMTECH	9
Skp2 inhibits osteogenesis by promoting ubiquitin–proteasome…	2016	CDRI	10
Drug Repurposing: A new Front in the War Against…	2016	CDRI	4
Globular adiponectin reverses osteo-sarcopenia and altered…	2017	CDRI	28

Indian citation flow

Citations received from (in) and given to (out) works carrying another Indian institution



Citations received per citing year (2026 partial)*

0.87	1.07	1.05	0.79
CCMB in/out ratio	IGIB in/out ratio	IMTECH in/out ratio	CDRI in/out ratio

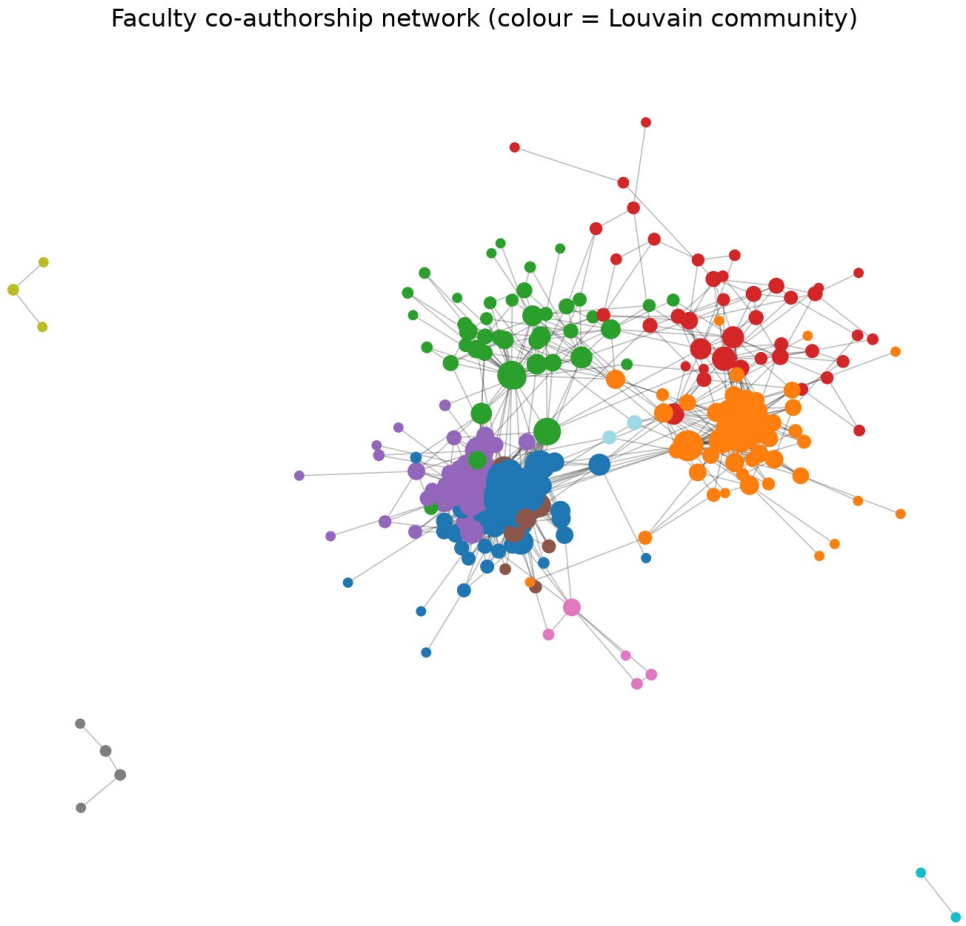
Top Indian institutes citing each lab

Lab	Citing institute	Cites in
CCMB	Indian Institute of Chemical Technology	318
	NIPER (Hyderabad)	221
	National Chemical Laboratory	192
IGIB	All India Institute of Medical Sciences	711
	University of Delhi	505
	Maharshi Dayanand University	343
IMTECH	IIIT Delhi	327
	Panjab University	260
	NIPER (Mohali)	167
CDRI	Jamia Hamdard	545
	SRM Institute of Science and Technology	496
	NIPER (Hyderabad)	449

In = CCMB 5,400, IGIB 7,569, IMTECH 4,806, CDRI 13,127; out = CCMB 6,228, IGIB 7,046, IMTECH 4,567, CDRI 16,565

Authorship network: brokers and communities

Faculty co-authorship graph: 274 faculty, 1311 links, Louvain modularity Q = 0.54



Most collaborative (degree)

Faculty	Lab	Deg
Kalyan Mitra	CDRI	51
Jiaur R Gayen	CDRI	47
Rabi S Bhatta	CDRI	45
Mohammad I Siddiqi	CDRI	44
Madhav N Mugale	CDRI	43

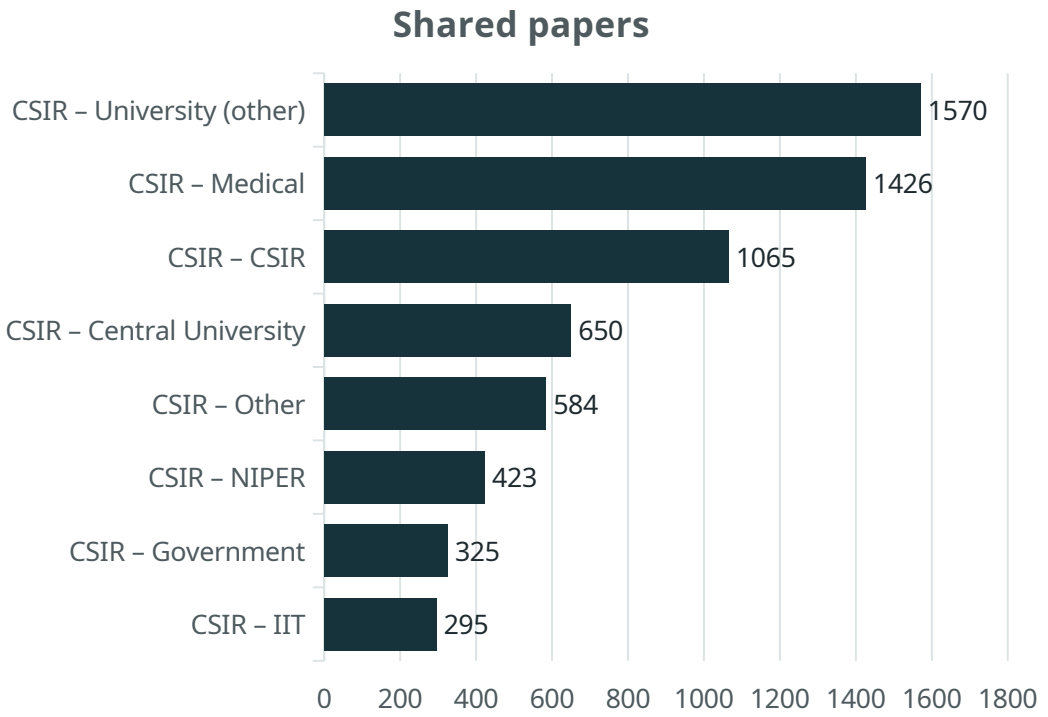
Bridges (betweenness)

Faculty	Lab	Betw
Rajesh Pandey	IGIB	0.165
Deepak Sharma	IMTECH	0.095
P. Anil Kumar	IMTECH	0.089
Divya Tej Sowpati	CCMB	0.068
Krishan Gopal	IMTECH	0.052

- 31 communities; the largest have 54, 51 and 43 faculty; 44% of communities with 3+ members sit within a single lab
- 98 of 1311 faculty links join two labs; 20 faculty have no faculty co-author
- Hub author: Shantanu Sengupta (IGIB) co-authors with 116 Indian partner institutions
- All-author graph: 6,124 authors, Q = 0.76; 48 papers with > 50 authors excluded from pairs

Connectivity map and key findings

Who our labs co-author with across Indian science



Shared papers by partner cluster (all four labs)

Lab	Top partner	Papers	Jaccard
CCMB	CSIR-IICT	106	0.08
IGIB	AIIMS	250	0.10
IMTECH	Panjab University	83	0.14
CDRI	King George's Medical University	138	0.04

Key findings

- CDRI is the most productive lab (2,699 papers); IMTECH has the highest Q1 share (76%).
- 56% of papers have an Indian partner co-author and 21% a foreign one.
- Strongest ties outside CSIR: University (other) (1,570 papers), Medical (1,426 papers), Central University (650 papers).
- Indian citation in/out ratio 0.79–1.07: IGIB, IMTECH receive more than they give; CCMB, CDRI give more than they receive.
- Collaboration is lab-bound: 98 of 1311 faculty links and 5% of lab-to-lab citations cross between the four labs.